Understanding Basic Probability Distributions

This report explains three common ways to look at data and probability. We will use simple charts to understand how data is spread out, how likely certain events are, and how to keep a running total of those chances.

1. The Normal Curve and the Empirical Rule

The first chart is called a "Bell Curve" or Normal Distribution. It shows that most data points happen near the middle (the average). As you move further away from the middle, in either direction, occurrences become much rarer.

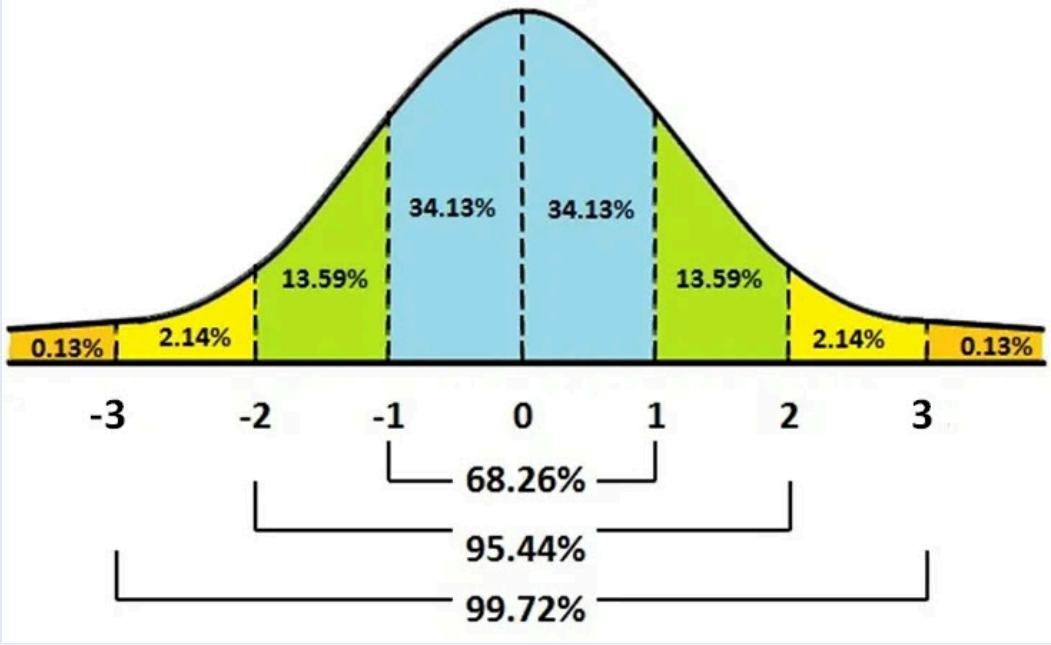


Figure 1: The 68-95-99.7 Rule applied to a normal distribution.

Statisticians measure the distance from the middle in "steps" called standard deviations. There is a helpful shortcut called the **68-95-99.7 Rule**:

Distance from the Middle	Amount of Data Inside this Area
1 Step (Standard Deviation)	About 68.26%
2 Steps	About 95.44%
3 Steps	About 99.72%

2. Finding the Area (Probability Density)

The second chart looks similar to the first, but it is used to calculate exact chances. In this chart, the total area under the entire blue curve is exactly 1 (which means 100% certainty).

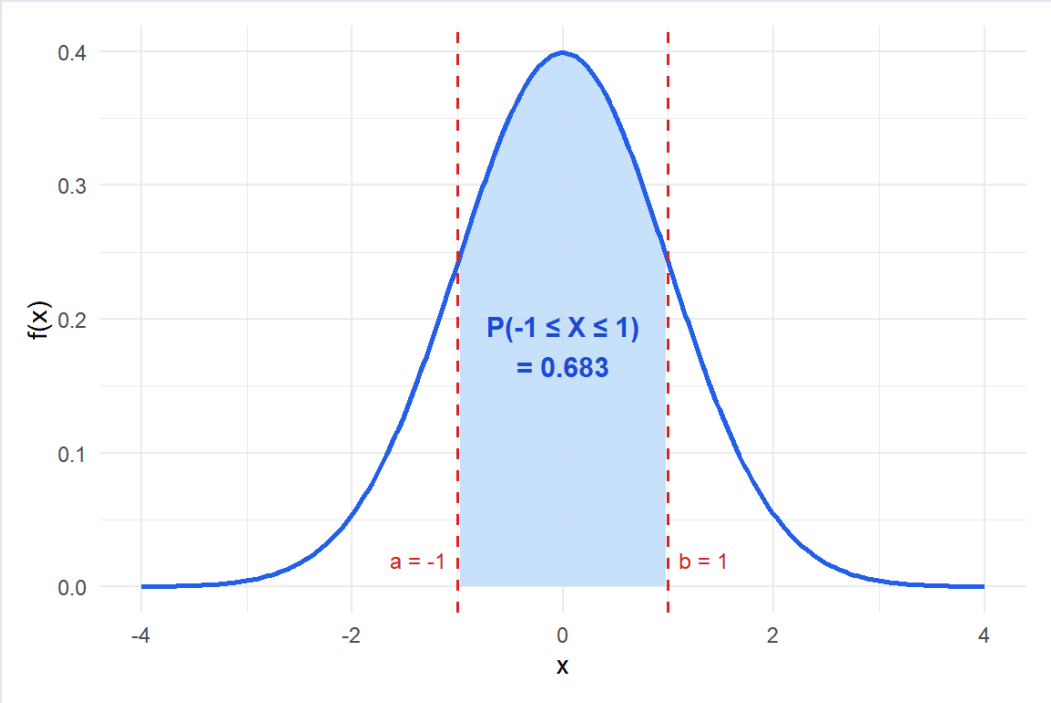


Figure 2: The shaded area represents the probability between -1 and 1.

If you want to know the chance of a number falling between two points, you just measure the area between them. The shaded blue part in Figure 2 is between the -1 and 1 marks. Just like our rule from Figure 1, the area here equals 0.683 (or 68.3%).

3. Keeping a Running Total (Cumulative Distribution)

The final chart is called a Cumulative Distribution Function (CDF). Instead of a bell shape, it looks like a staircase going up. It keeps a running total of probabilities from left to right.

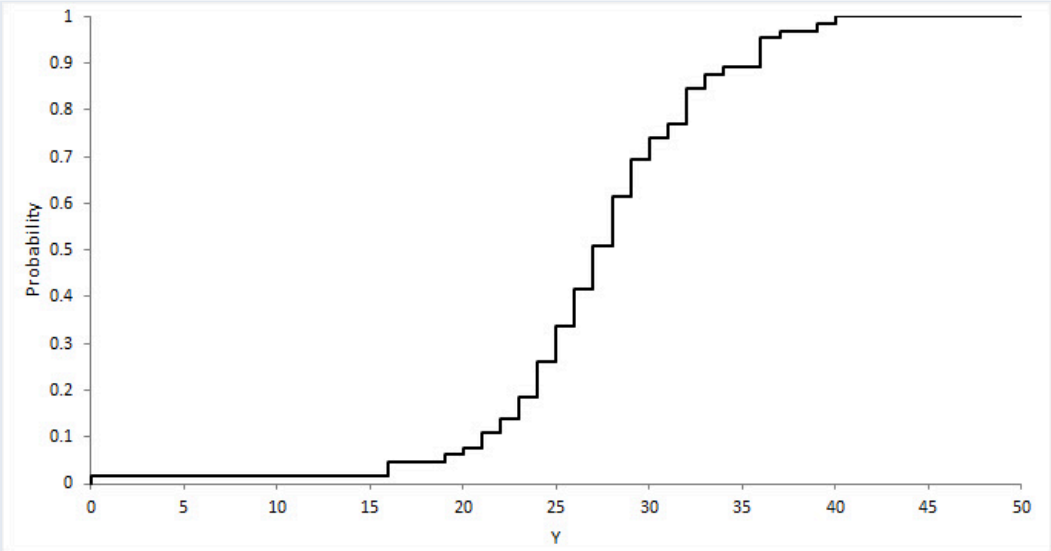


Figure 3: A step chart showing accumulated probability up to 100%.

The bottom of the chart starts at 0. As you move to the right, the line steps up every time it includes more data. It stops at 1 (or 100%) at the top right, meaning all possible data points have been counted. If you pick a point on the bottom line and look at how high the step is, it tells you the exact chance that a result will be *that number or lower*.